

Multi-seed M-finescan: vanilla vs orthogonal AL-RNN

Sweep: M in [10, 11, 14, 15] x P in [2, 3, 4, 5, 6] x seeds in [0, 1, 2] x {vanilla, ortho}

Epochs: 1000 alpha: 1.0 shared frozen B per M (seed_B = 42)

fixed Q_dataset rotation (seed_data = 42)

Per-seed seed_run controls model init + Python random batch sampling.

Metrics (lower is better):

rmse_train_final - sqrt(final training MSE), one-step prediction

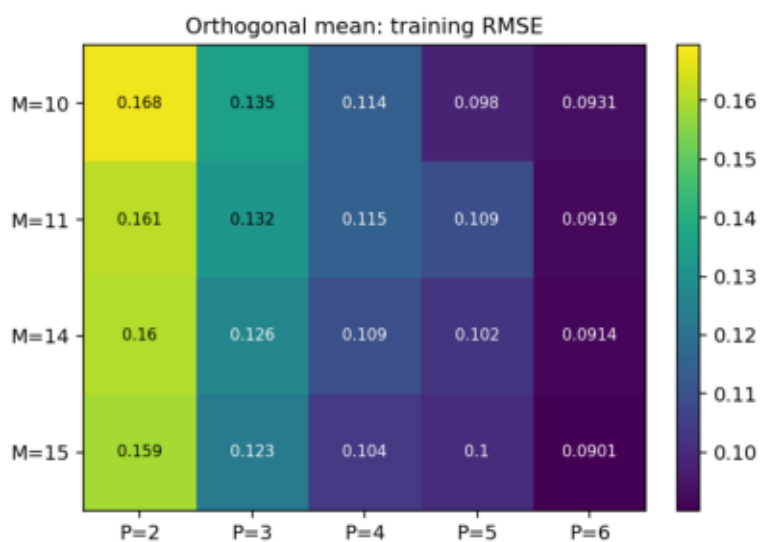
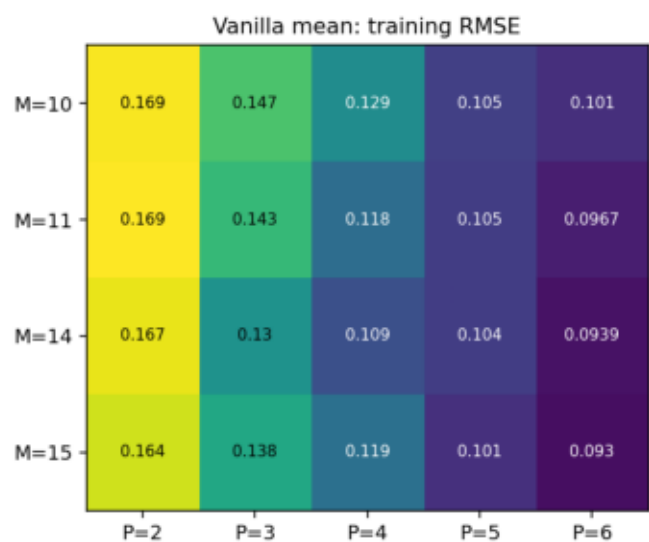
Dstsp - KL(p_true || p_gen) on a 30^3 binned 3-D density

PSE - mean Hellinger distance between smoothed power spectra

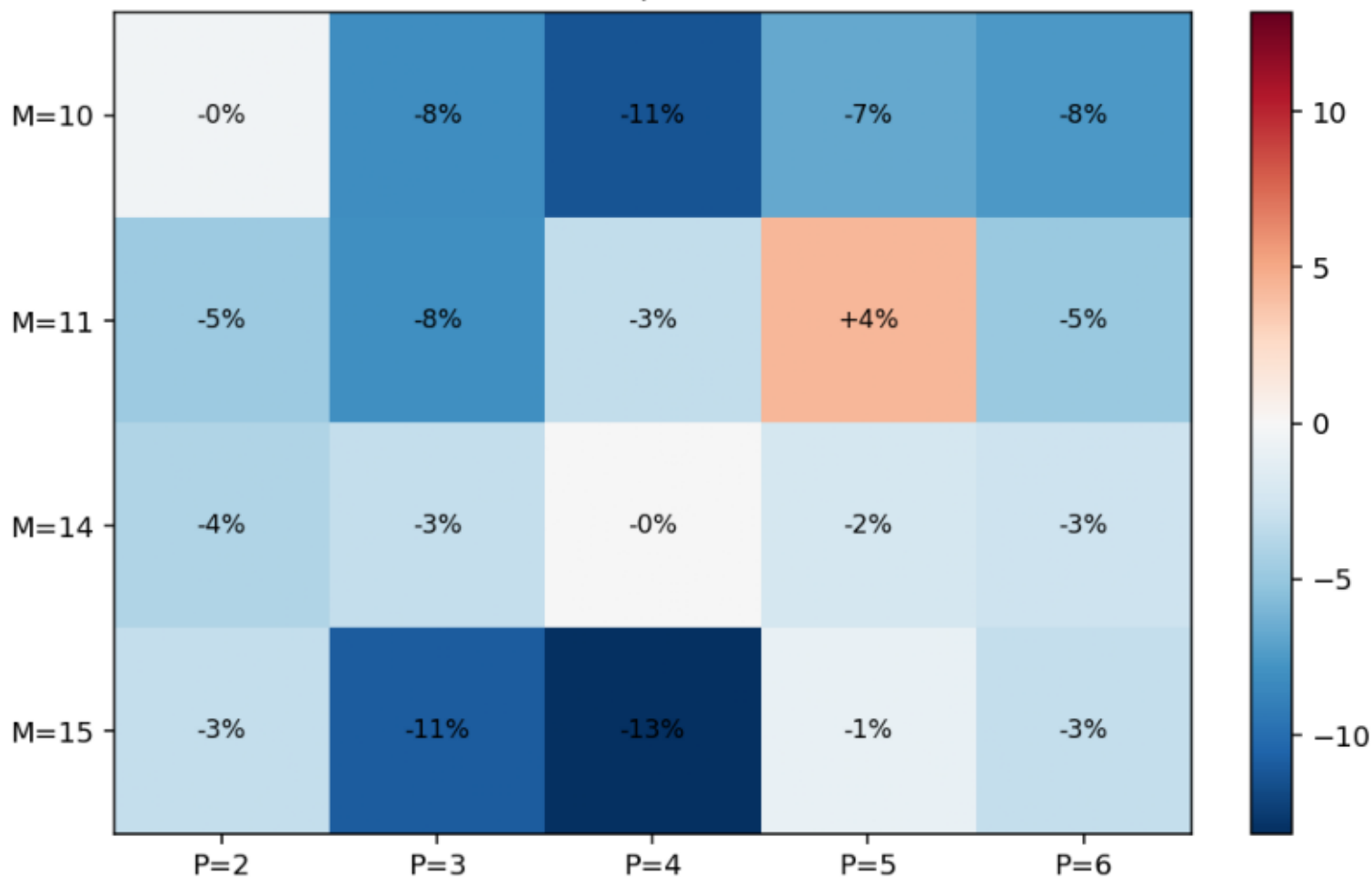
D_H_state - Hellinger distance on the same state-space density

Per-cell summary uses mean +/- std over the seeds; the 'win rate' heatmap shows the fraction of seeds where ortho < vanilla on that metric.

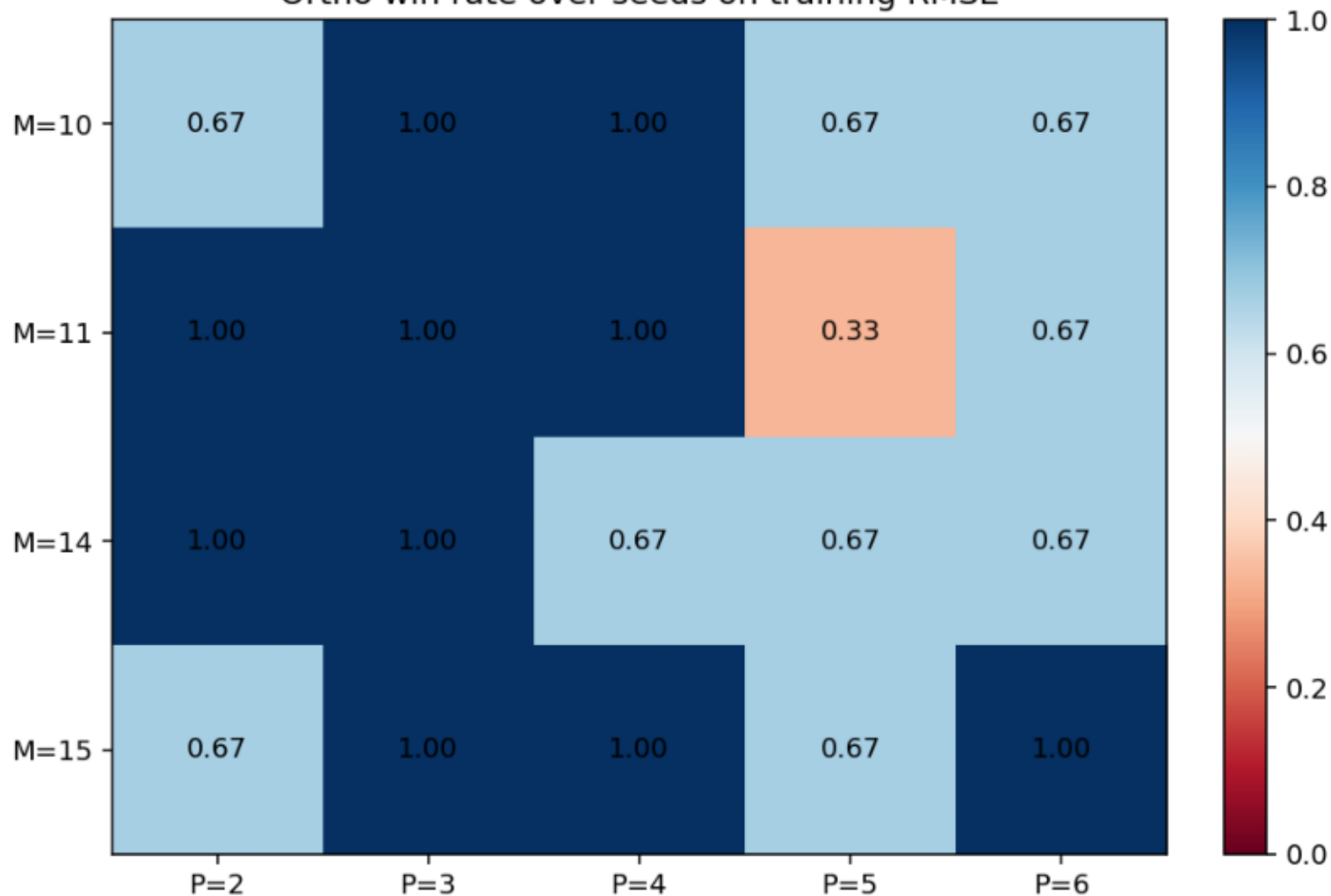
training RMSE (mean over 3 seeds, lower is better)



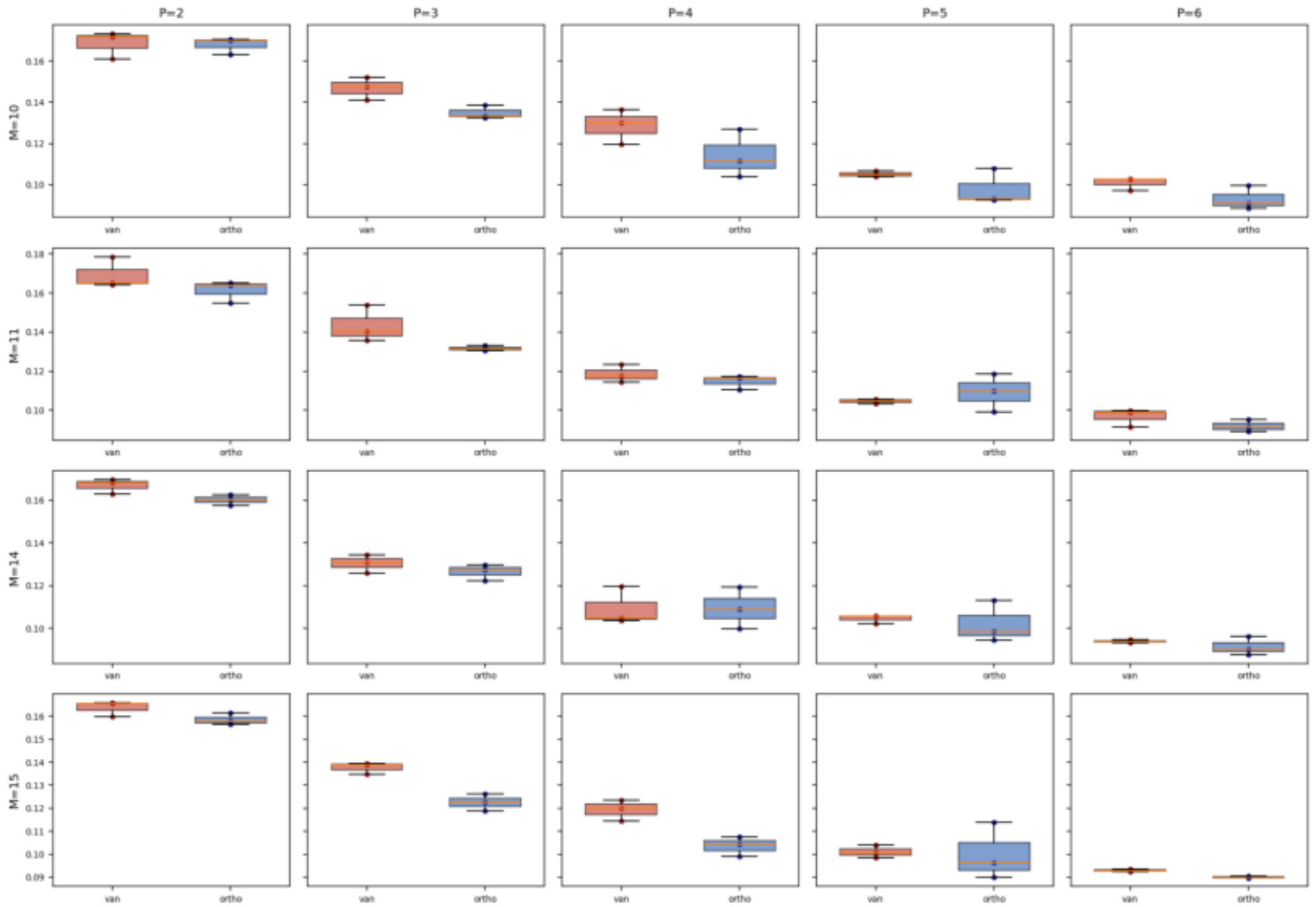
Gap on mean training RMSE
blue = ortho wins, red = vanilla wins



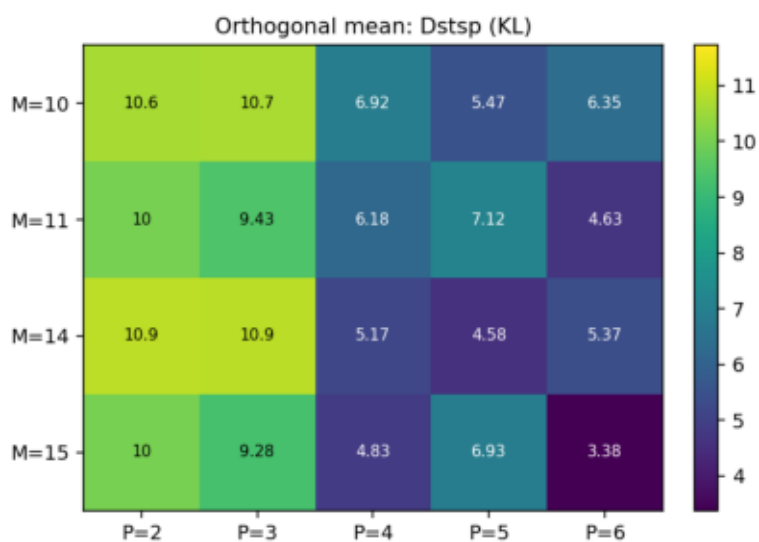
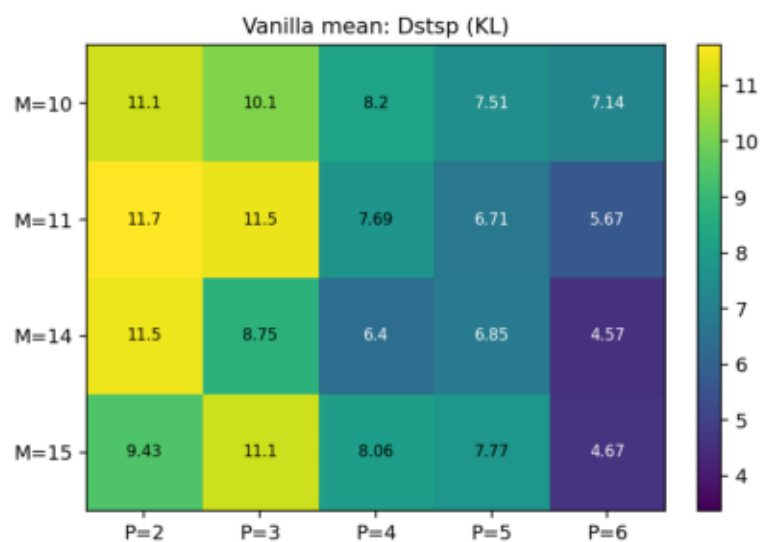
Ortho win rate over seeds on training RMSE



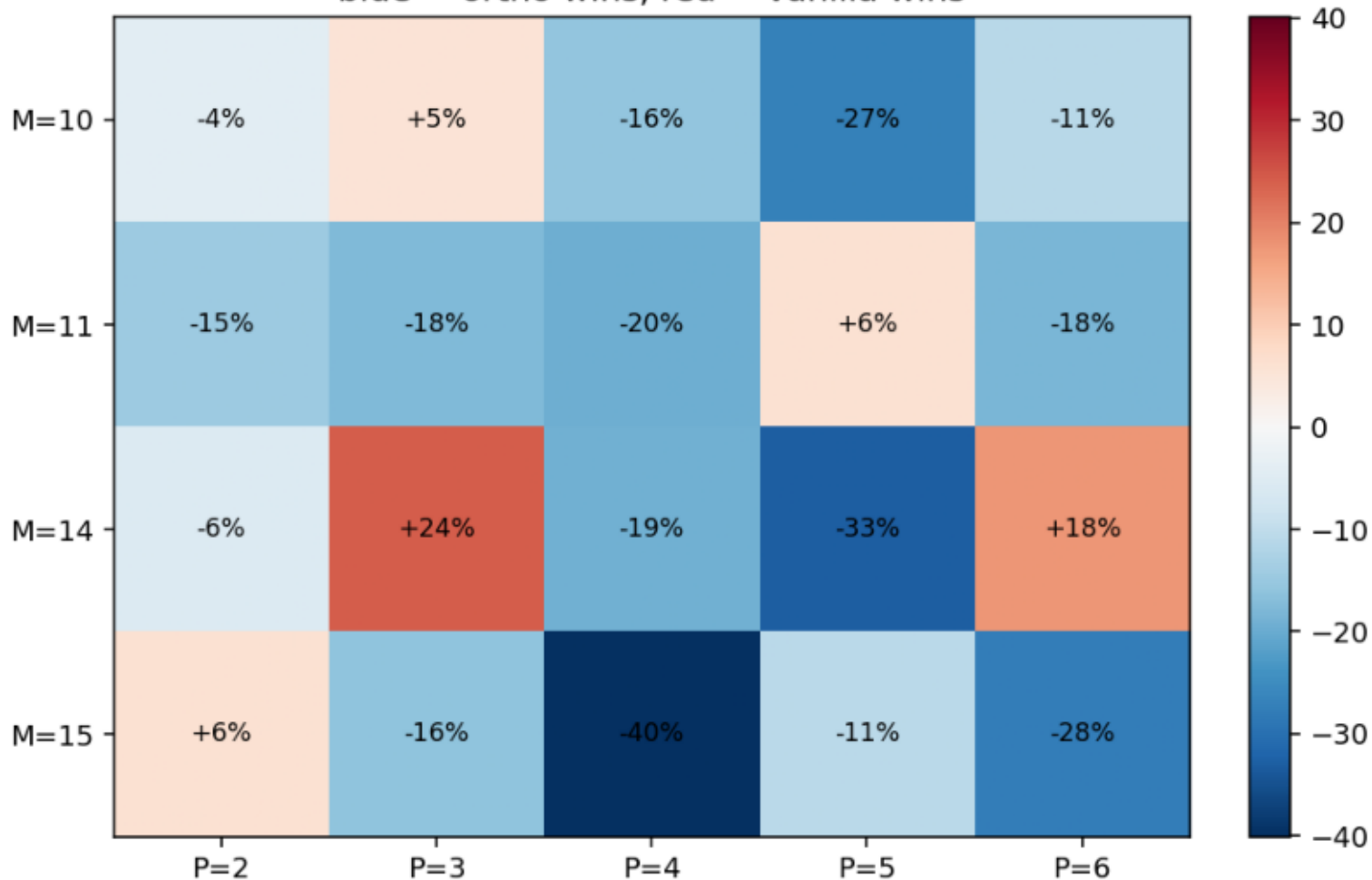
training RMSE (per-seed values; box = IQR, line = median)



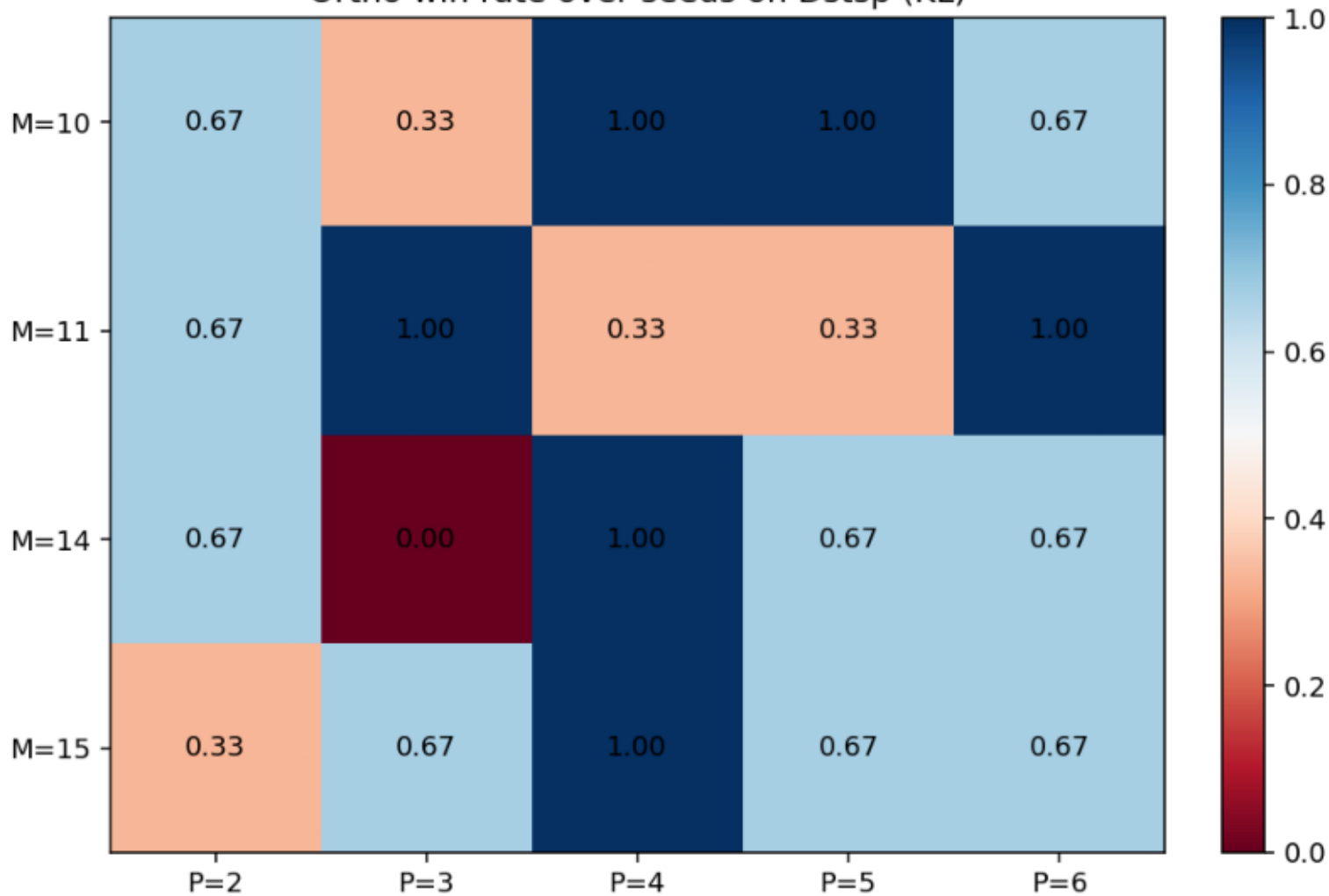
Dstsp (KL) (mean over 3 seeds, lower is better)



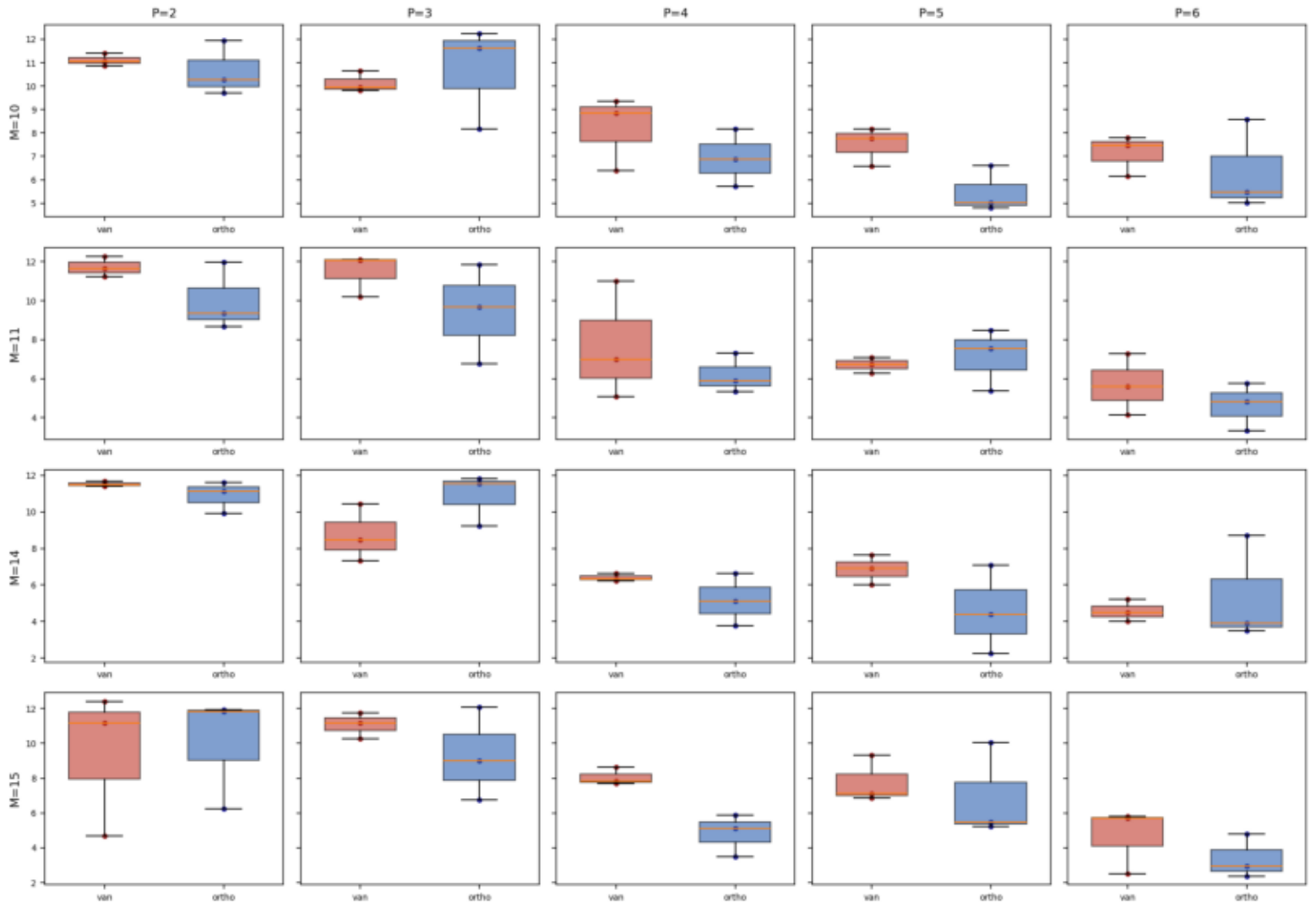
Gap on mean Dstsp (KL)
blue = ortho wins, red = vanilla wins



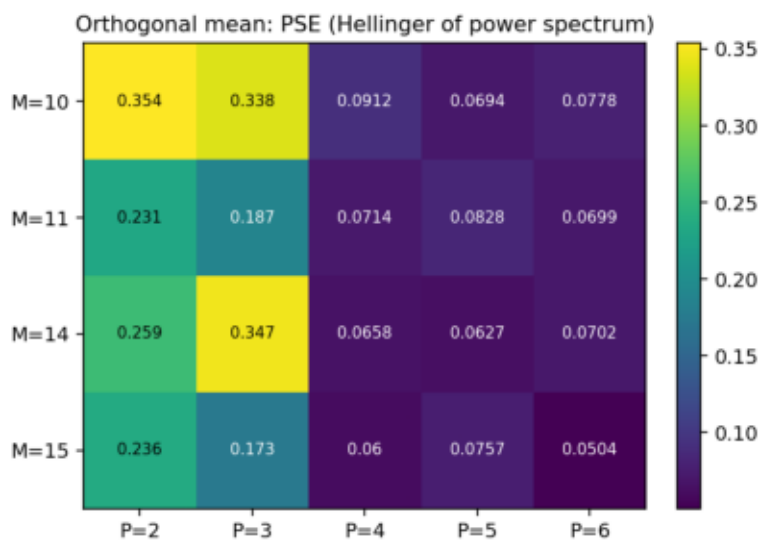
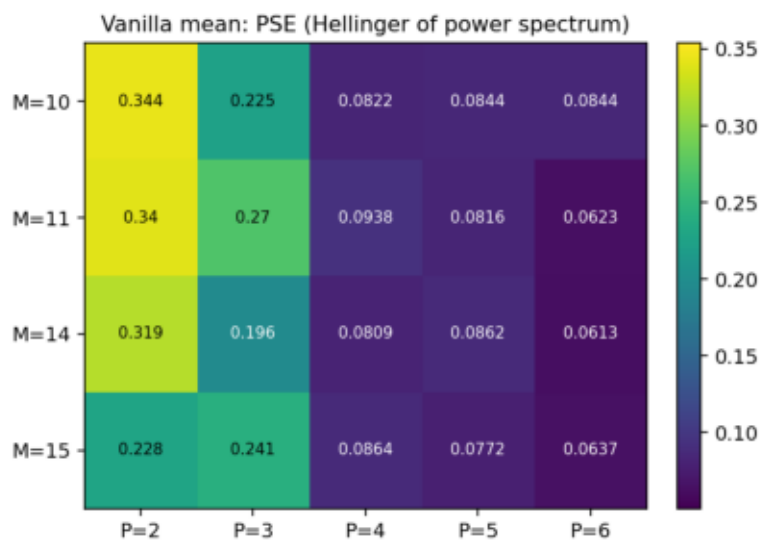
Ortho win rate over seeds on Dstsp (KL)



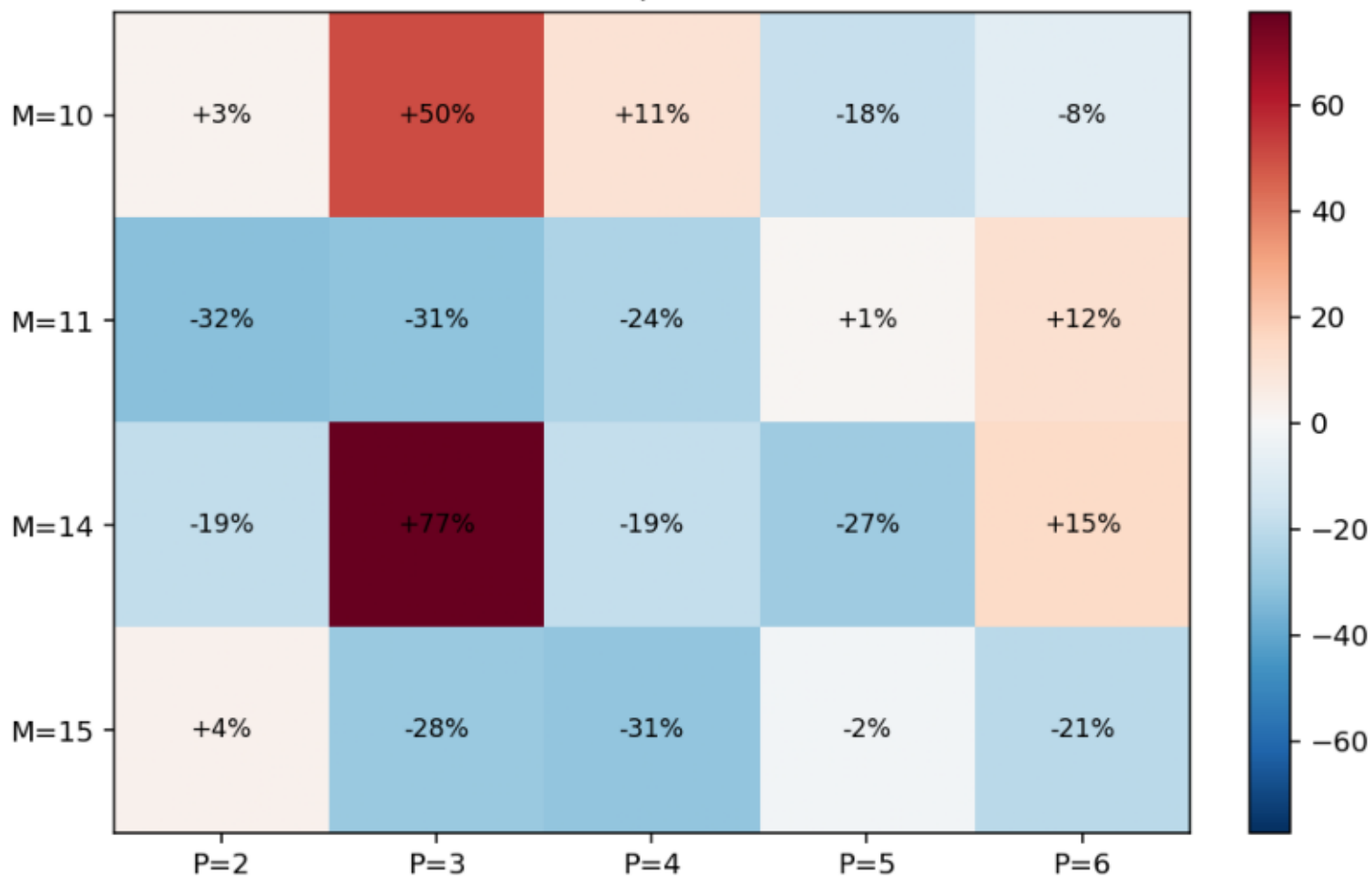
Dstsp (KL) (per-seed values; box = IQR, line = median)



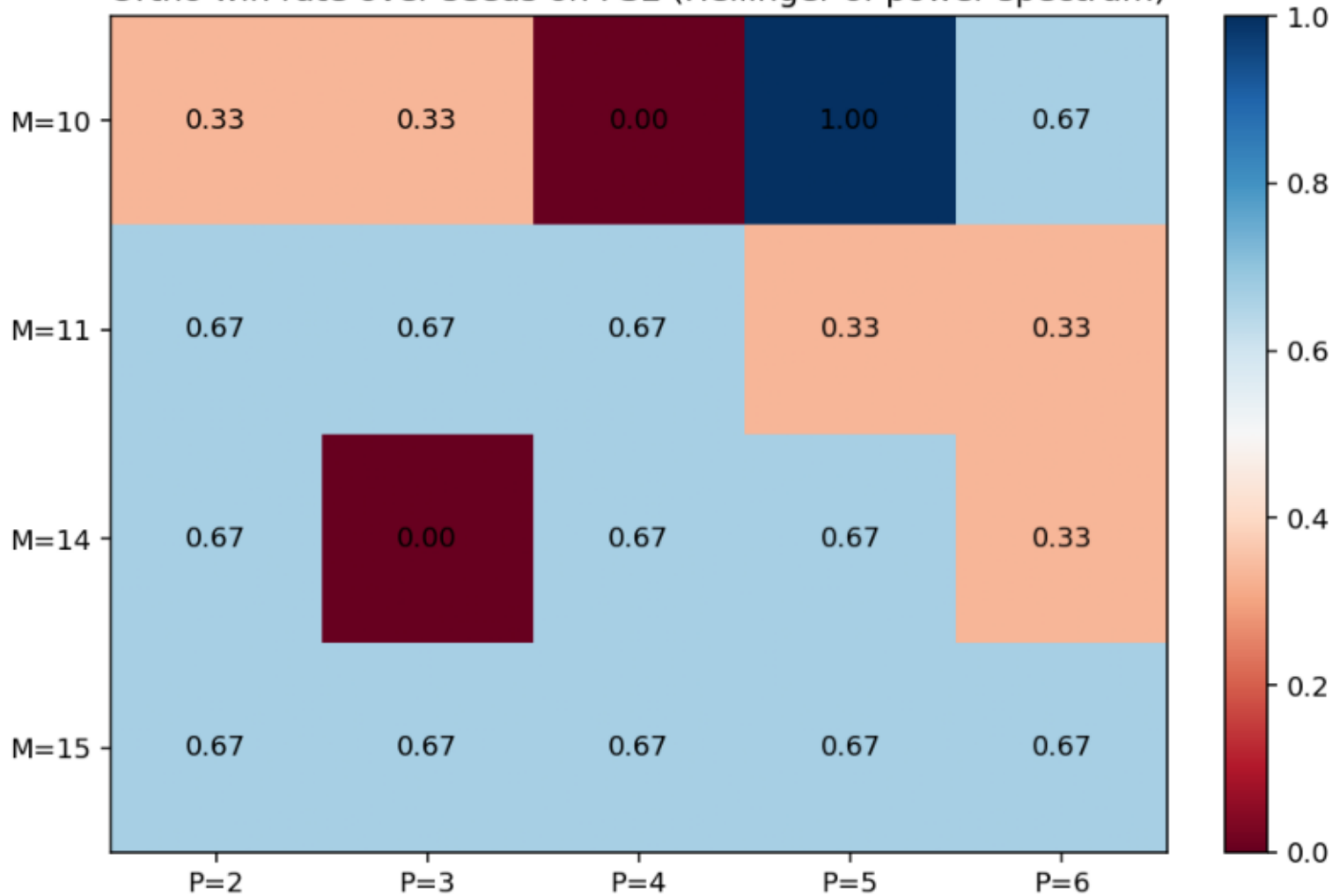
PSE (Hellinger of power spectrum) (mean over 3 seeds, lower is better)



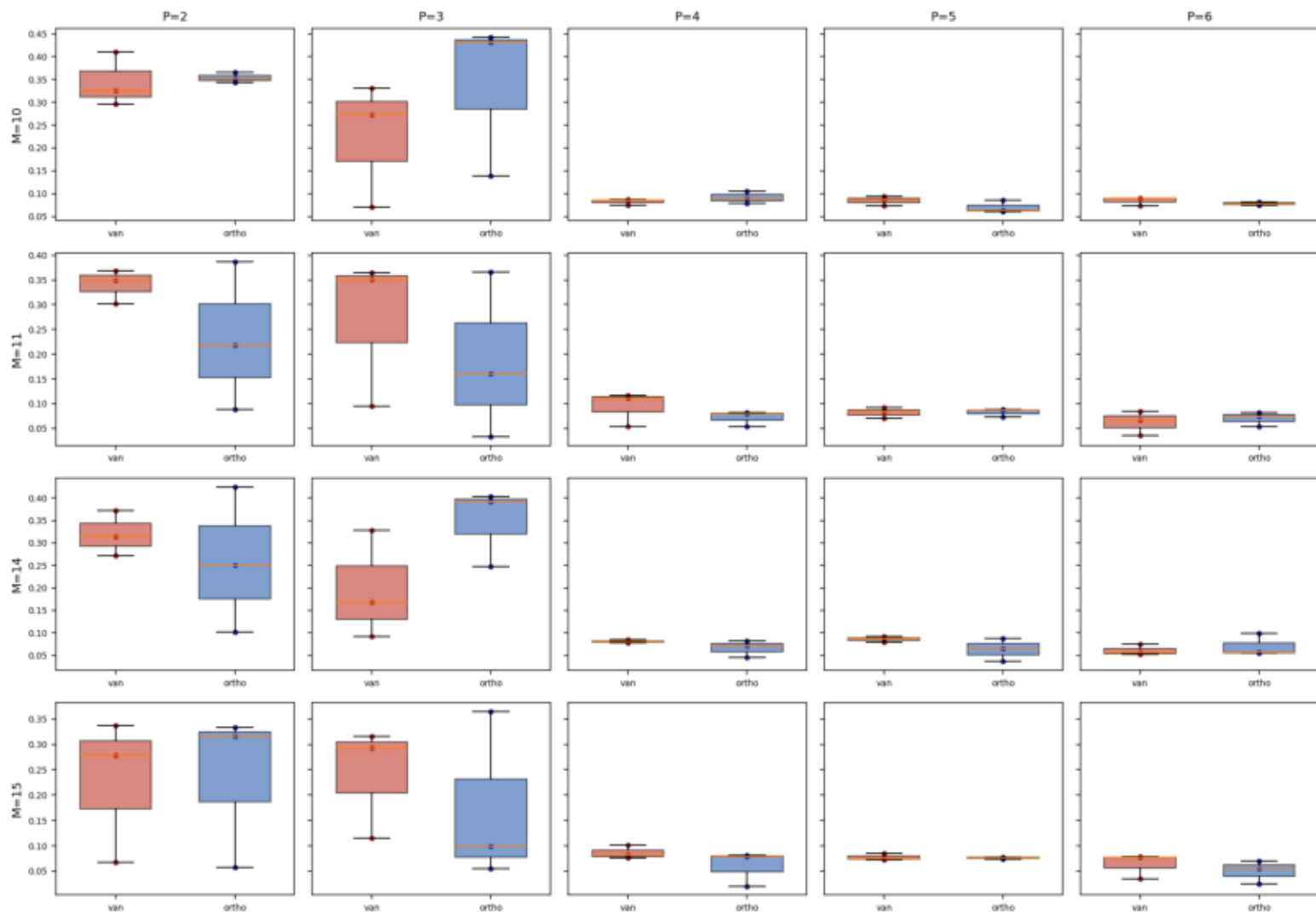
Gap on mean PSE (Hellinger of power spectrum)
blue = ortho wins, red = vanilla wins



Ortho win rate over seeds on PSE (Hellinger of power spectrum)

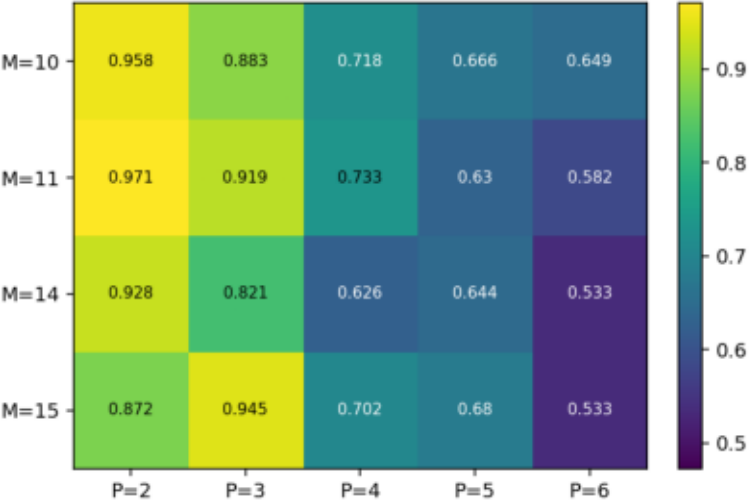


PSE (Hellinger of power spectrum) (per-seed values; box = IQR, line = median)

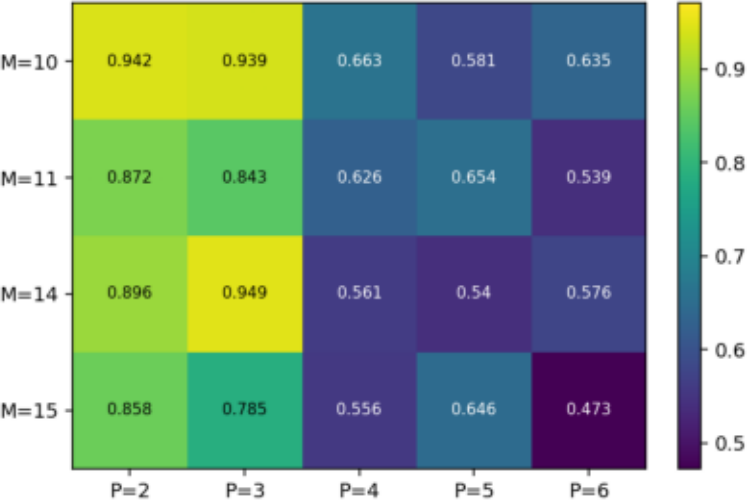


D_H state-space (Hellinger) (mean over 3 seeds, lower is better)

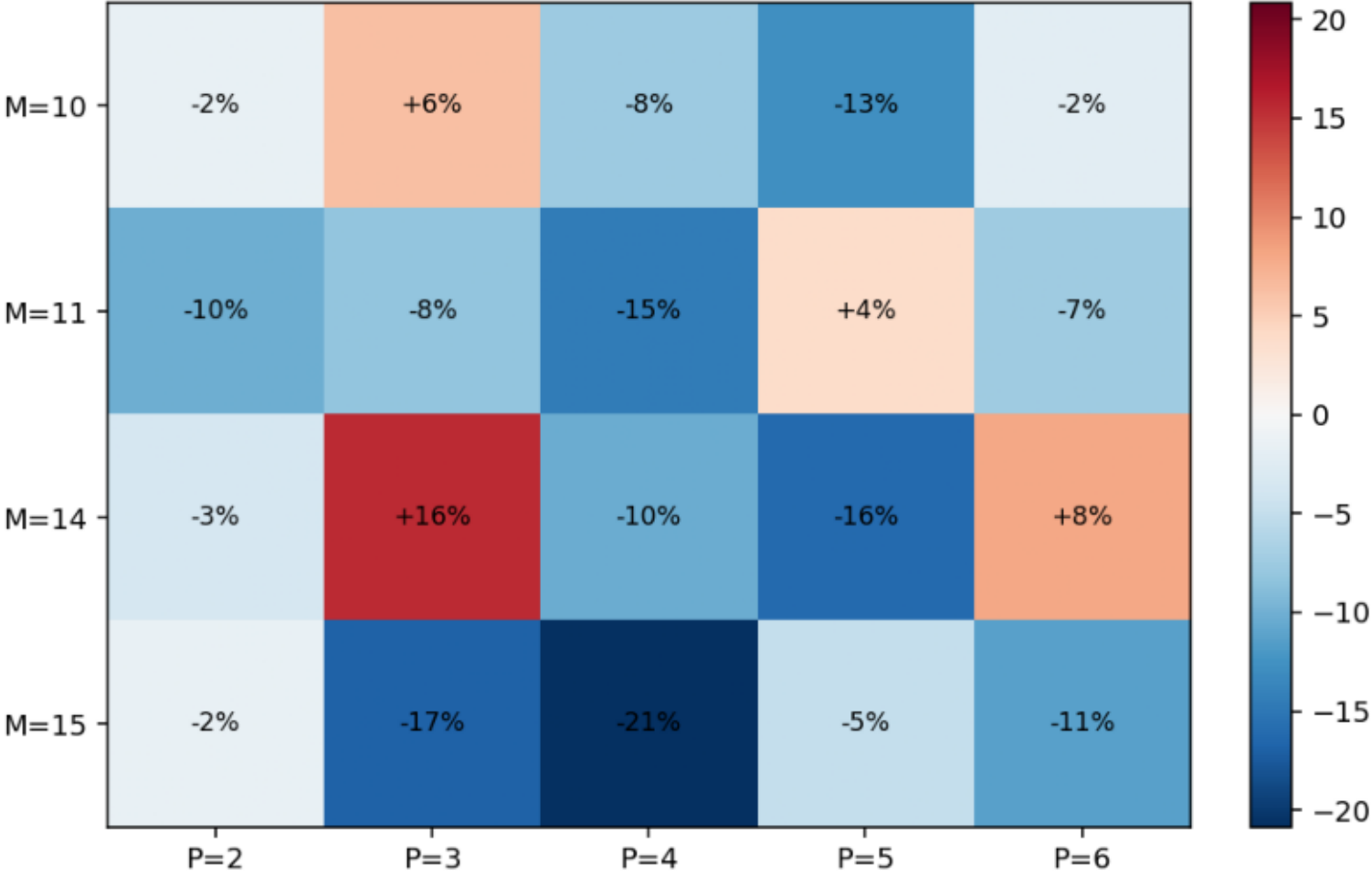
Vanilla mean: D_H state-space (Hellinger)



Orthogonal mean: D_H state-space (Hellinger)



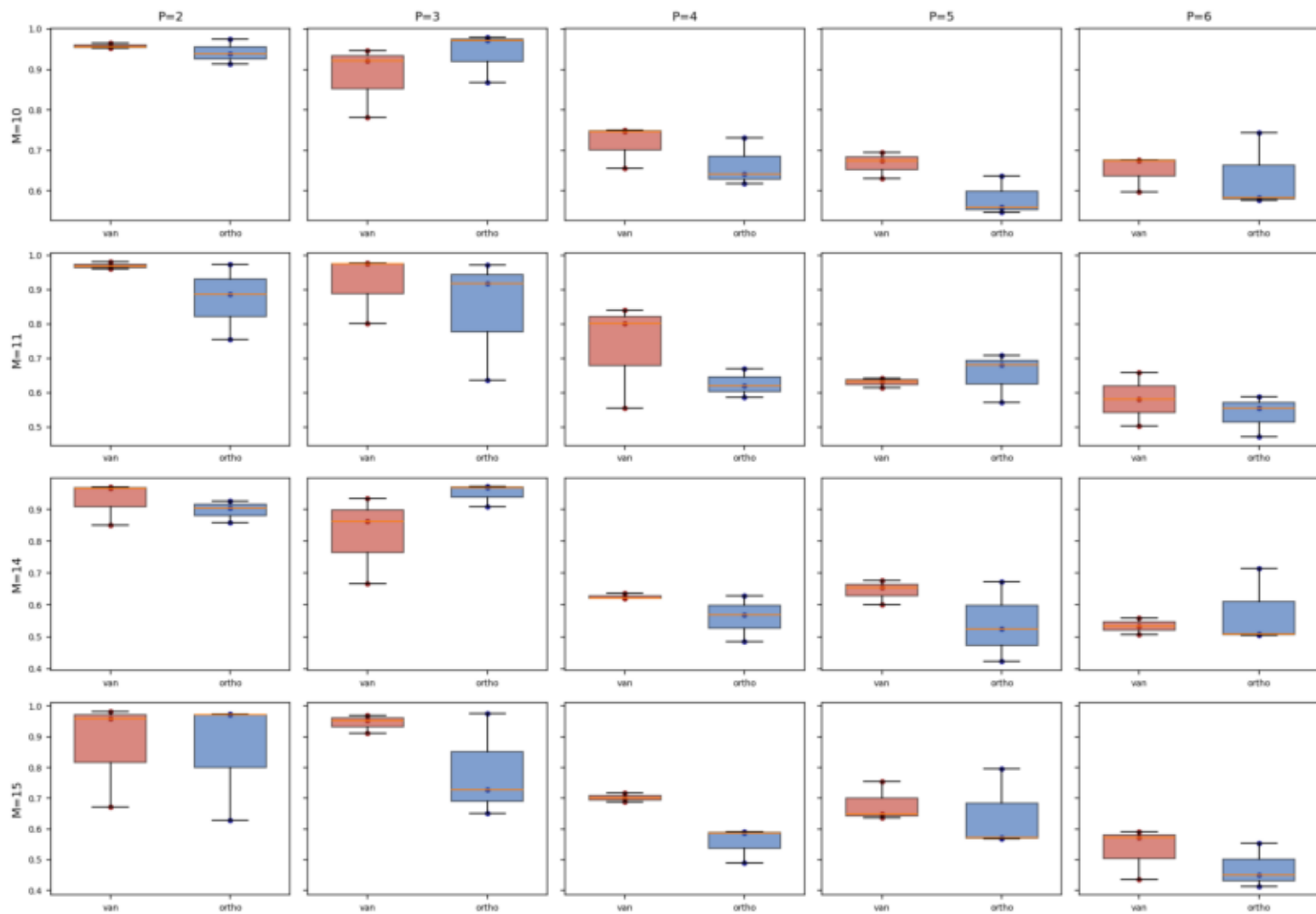
Gap on mean D_H state-space (Hellinger)
blue = ortho wins, red = vanilla wins



Ortho win rate over seeds on D_H state-space (Hellinger)



D_H state-space (Hellinger) (per-seed values; box = IQR, line = median)



Metrics vs M, mean +- std over 3 seeds (vanilla dashed, ortho solid)

